

pipeline_report

EP

2026-08-04

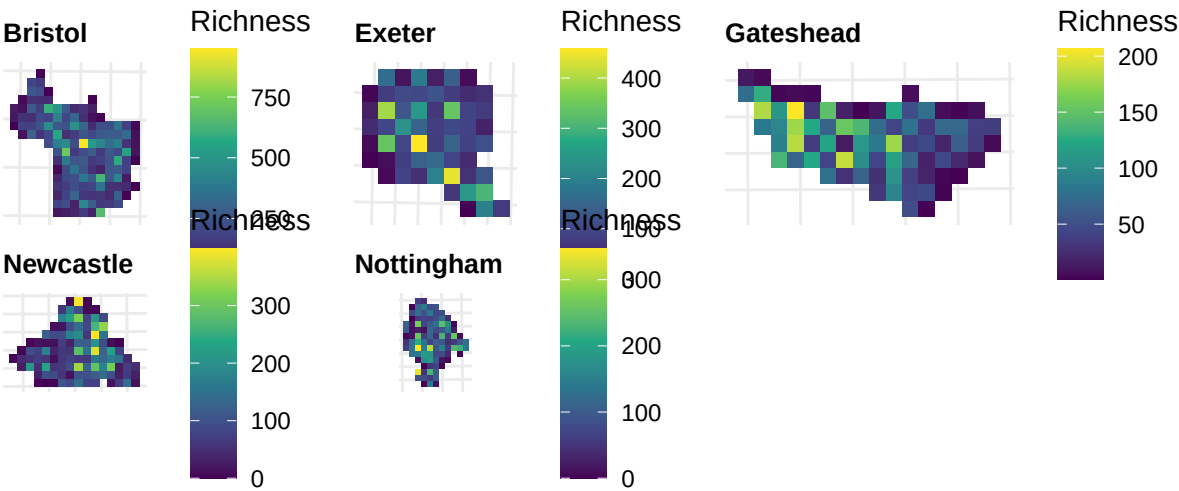
Overview

Table 1: Summary of diversity and recording effort by city

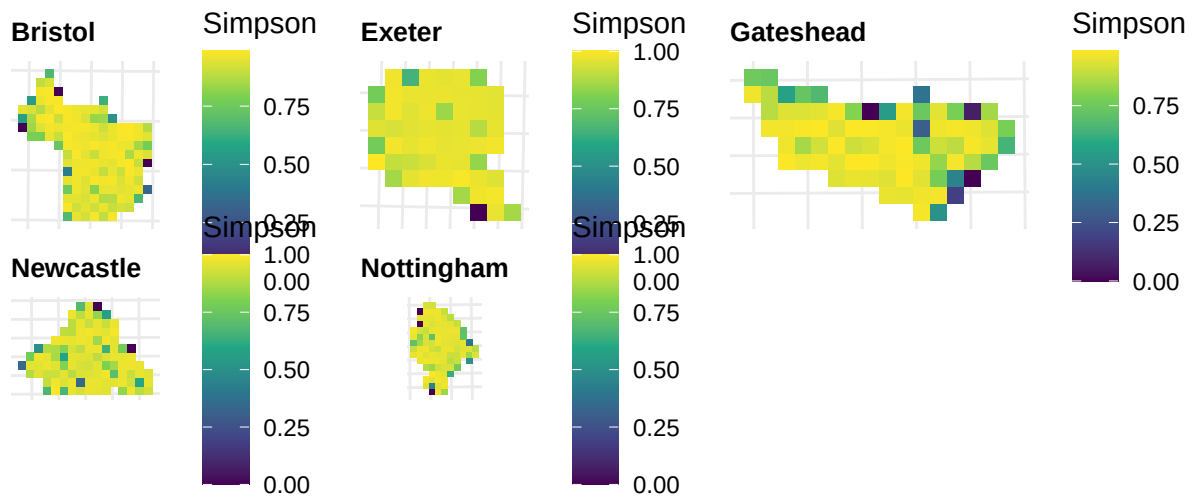
city	n_cells	mean_richness	mean_shannon	mean_simpson	total_records	total_survey_days
Bristol	143	218.08	3.39	0.89	836079	84948
Exeter	58	121.90	3.41	0.92	218942	21875
Gateshead	80	63.86	2.86	0.83	223262	19009
Newcastle	109	105.07	3.15	0.88	318004	35752
Nottingham	99	100.35	3.22	0.89	199674	27043

Spatial patterns

Species richness



Simpson diversity



Predictor summary

Table 2: Mean predictor values by city (terrain, imperviousness, forest, tree height)

city	dtm_elev	avg_imperv	avg_area_ha	treeht
Bristol	44.88	38.74	7.65	5.16
Exeter	39.32	25.08	2.13	5.28
Gateshead	58.25	34.54	3.92	4.35
Newcastle	59.38	37.33	3.53	4.63
Nottingham	53.12	39.02	4.36	5.58

Table 3: Mean predictor values by city (climate, river distance, greenspace)

city	weighted_precip	weighted_temp	dist_to_river	pct_greenspace
Bristol	878.61	10.97	622.08	0.15
Exeter	890.47	10.77	405.81	0.07
Gateshead	713.29	9.61	755.25	0.12

city	weighted_precip	weighted_temp	dist_to_river	pct_greenspace
Newcastle	688.83	9.56	893.98	0.14
Nottingham	682.74	10.40	814.21	0.12

Table 4: Cell counts by surface water flood risk (RoFSW) category and city

rofsw_risk	Bristol	Exeter	Gateshead	Newcastle	Nottingham
Low	94	51	68	93	94
Medium	19	0	0	0	1
High	30	6	0	14	3
NA	0	1	12	2	1

Table 5: Cell counts by river and sea flood risk (RoFRS) category and city

rofrs_risk	Bristol	Exeter	Gateshead	Newcastle	Nottingham
Very low	17	0	1	1	3
Low	24	14	11	13	28
Medium	3	2	4	1	4
High	65	29	32	48	27
NA	34	13	32	46	37

Models

Species richness (negative binomial GLM)

Table 6: Full richness model coefficients

term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.171	1.022	0.167	0.867
n_days	0.001	0.000	10.736	0.000
dtm_elev	0.000	0.002	0.133	0.894
avg_imperv	0.005	0.003	1.582	0.114
rofsw_riskMedium	0.108	0.259	0.419	0.675
rofsw_riskHigh	0.163	0.139	1.171	0.242
rofrs_riskLow	-0.149	0.257	-0.579	0.563
rofrs_riskMedium	-0.056	0.330	-0.170	0.865
rofrs_riskHigh	-0.209	0.250	-0.838	0.402
avg_area_ha	0.003	0.004	0.778	0.437
treht	-0.027	0.040	-0.670	0.503
weighted_precip	0.000	0.001	0.452	0.651
weighted_temp	0.374	0.137	2.723	0.006
dist_to_river	0.000	0.000	-0.618	0.536
pct_greenspace	1.734	0.360	4.816	0.000

Table 7: Richness: Null vs. Full models

logLik_null	logLik_full	AIC_null	AIC_full	chisq	df	p_value
-2879.981	-1788.753	5763.963	3609.505	2182.458	14	0

Simpson diversity (beta regression)

Table 8: Full Simpson model coefficients

term	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.988	1.328	0.744	0.457
n_days	0.000	0.000	1.774	0.076
dtm_elev	-0.001	0.003	-0.196	0.845
avg_imperv	0.004	0.004	0.942	0.346
rofsw_riskMedium	0.309	0.337	0.915	0.360
rofsw_riskHigh	0.119	0.181	0.655	0.512
rofrs_riskLow	0.395	0.333	1.185	0.236
rofrs_riskMedium	0.494	0.430	1.149	0.251
rofrs_riskHigh	0.288	0.324	0.891	0.373
avg_area_ha	0.003	0.005	0.564	0.573
treeht	0.030	0.052	0.567	0.571
weighted_precip	0.000	0.001	-0.062	0.950
weighted_temp	-0.020	0.178	-0.115	0.909
dist_to_river	0.000	0.000	-0.685	0.494
pct_greenspace	1.019	0.471	2.163	0.031

Table 9: Simpson: Null vs. Full Models

logLik_null	logLik_full	AIC_null	AIC_full	chisq	df	p_value
396.25	277.201	-788.499	-522.402	-238.097	14	1

Recording effort

